SUPPORTING INFORMATION

STUDY ON LATENT-SPACE TRAVERSAL

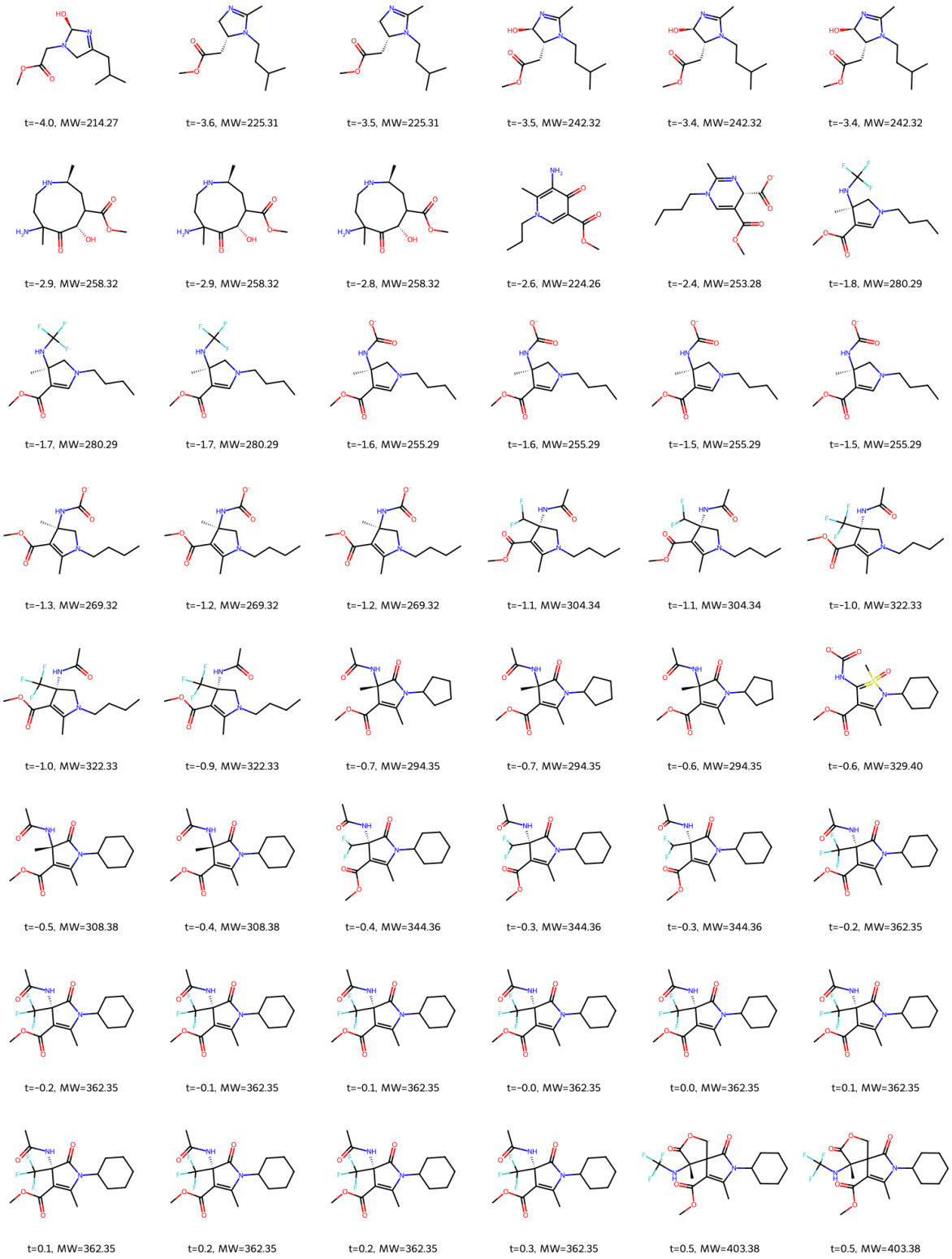


Figure S1. Latent-space traversal along a learned molecular weight (MW) direction. Starting from an anchor molecule (center), latent vectors were incrementally perturbed using a traversal parameter *t* (negative to positive steps along the MW direction) and decoded to generate chemically valid structures. The resulting molecules exhibit smooth and systematic changes in molecular size while preserving core chemical motifs, illustrating the continuity and property-aware organization of the learned latent manifold.

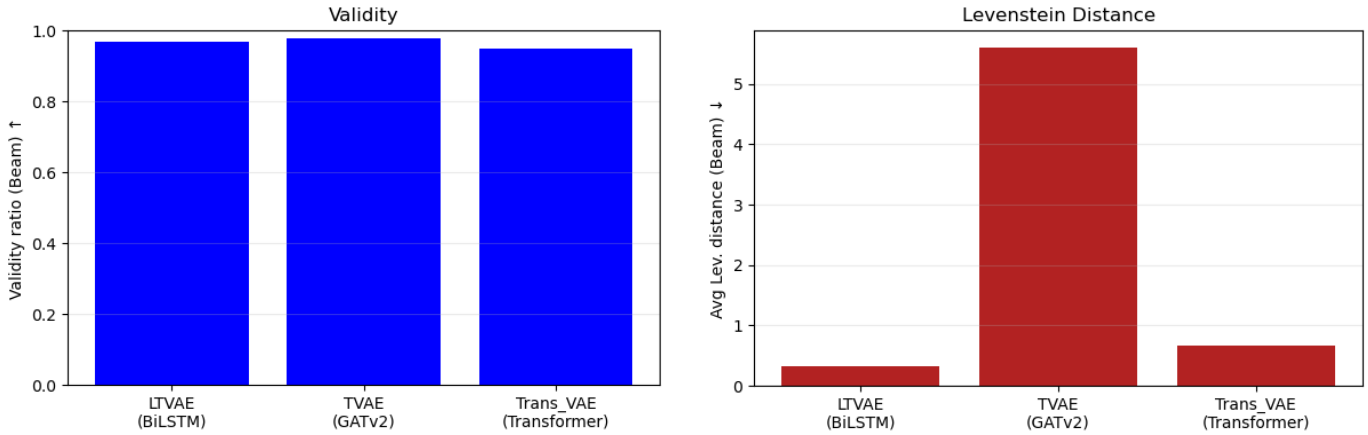
To further probe the structure and interpretability of the learned latent space, latent-space traversal experiments were performed along a direction corresponding to molecular weight (MW). This direction was identified by fitting a linear model between latent embeddings and RDKit-computed MW values over a representative subset of molecules, yielding a continuous vector in latent space associated with monotonic changes in molecular size. Starting from an anchor molecule, the latent representation was incrementally perturbed along this learned MW direction using a scalar traversal parameter t, with negative and positive values corresponding to decreasing and increasing molecular weight, respectively. Each perturbed latent vector was then decoded to generate a sequence of candidate molecular structures.

As illustrated in Figure S1, the decoded molecules exhibit smooth and systematic changes in molecular size as t varies, while largely preserving core chemical scaffolds and functional motifs. Importantly, the majority of generated structures remain chemically valid across the traversal range, and the observed changes in MW follow a consistent monotonic trend. These results indicate that the learned latent manifold is not only globally correlated with molecular properties but also locally continuous and semantically organized, enabling controlled exploration of chemically meaningful variations. Such behavior supports the suitability of the proposed latent representation for property-aware molecular manipulation and downstream applications requiring interpretable structure–property relationships.

ABLATION STUDY

1. Encoder Ablation

To assess the influence of the encoder architecture on model performance, an encoder ablation study was conducted in which only the encoder component of the variational autoencoder was varied, while all other aspects of the model were held constant. Three encoder types were considered: a bidirectional LSTM operating on tokenized SMILES strings (LTVAE), a graph-based encoder employing GATv2 layers applied to molecular graphs (TVAE) [1], and a Transformer encoder applied to SMILES sequences (Trans_VAE) [2]. In all cases, the decoder architecture, latent dimensionality, training dataset, optimization schedule, and decoding strategy were identical.



(a) (b)

Figure S2. Encoder ablation results. Beam-search reconstruction performance for different encoder architectures. (a) chemical validity ratio of reconstructed SMILES, and (b) average Levenstein distance between reconstructed and target SMILES. All models use the same decoder, latent dimensionality, beam settings, and training protocol; only the encoder architecture is varied.

Table S1: Encoder ablation study

Model	Encoder Type	Parameters	Val Acc. (TF)	Val Acc. (Beam)	Validity (Beam)	Avg. Lev. Dist. (Beam)
TVAE [1]	GATv2	3.72M	96.60	78.90	98.10	5.60
Trans_VAE [2]	Transformer	5.54M	99.20	96.90	95.70	0.66
LTVAE	BiLSTM	2.78M	99.60	98.10	97.80	0.33

The results, summarized in Table S1 and Figure S2, indicate that the choice of encoder has a measurable impact on reconstruction quality. The graph-based encoder yields a higher fraction of chemically valid reconstructions and a lower average Levenstein distance relative to the sequence-based encoders. This behavior suggests that incorporating explicit molecular graph information in the encoder facilitates the learning of latent representations that better preserve chemically relevant structural features. The Transformer encoder exhibits intermediate performance between the BiLSTM and graph-based encoders, indicating that increased sequence modeling capacity alone does not fully compensate for the absence of explicit structural information.

2. Bottleneck Ablation: Effect of Latent Dimensionality

The effect of the variational bottleneck on reconstruction fidelity and latent expressiveness was examined by systematically varying the latent dimensionality of the proposed LTVAE architecture. This analysis is summarized in Figure 5 of the main manuscript, where reconstruction accuracy, chemical validity ratio, and average Levenstein distance are reported across training epochs for latent dimensions of 4, 8, 16, 32, 64, 128, and 256. All experiments were conducted using the same encoder–decoder architecture, dataset, and training protocol, ensuring that observed performance differences arise solely from changes in bottleneck capacity.

As shown in Figure 5, very small latent spaces (latent dimension ≤ 16) exhibit limited representational capacity, resulting in substantially lower reconstruction accuracy and persistently higher structural deviation. A pronounced transition is observed between latent dimensions of 16 and 32, where reconstruction accuracy improves sharply and Levenstein distance decreases rapidly. This behavior indicates that this range represents the minimum latent capacity required to reliably encode chemically meaningful structure. For larger latent dimensionalities (≥ 64), reconstruction accuracy and validity approach saturation, while further increases in latent size provide only marginal improvement, reflecting diminishing returns in representational scaling.

Overall, this bottleneck ablation demonstrates that the proposed LTVAE framework preserves sufficient molecular information even with compact latent representations, and that a moderate latent dimensionality (e.g., 64) provides an effective balance between reconstruction fidelity, latent compactness, and computational efficiency. These results directly support the architectural design choices adopted in the main manuscript.

3. Ablation on Randomized SMILES & Token Corruption

To assess the impact of SMILES randomization and input-level regularization on model robustness, an ablation study was performed comparing two training configurations of the proposed LTVAE architecture. In the first configuration, encoder-side randomized SMILES augmentation was combined with decoder-side token corruption and label smoothing. In the second configuration, all three mechanisms were disabled, and the model was trained exclusively on canonical SMILES without input noise.

As shown in Figure S3 models trained without randomization exhibit faster convergence and marginally improved reconstruction performance on in-distribution ZINC15-like data, achieving slightly higher token-level accuracy and lower average Levenstein distance. This behavior reflects tighter memorization of canonical SMILES representations when no stochastic augmentation is applied.

In contrast, evaluation on the dye SMILES test set reveals a consistent advantage for the randomized training configuration. Models trained with SMILES randomization and token corruption achieve higher reconstruction accuracy, improved chemical validity, and reduced structural deviation relative to their non-randomized counterparts. These gains indicate enhanced robustness to SMILES ordering and improved generalization to chemically distinct molecules that are sparsely represented in large-scale drug-like datasets.

Overall, the ablation results highlight a trade-off between optimal in-distribution reconstruction and out-of-distribution generalization. While noise-free training favors ZINC-like molecules, the inclusion of randomized SMILES and token-level

corruption produces a more stable and transferable latent representation. Given the target application to specialized dye chemistries, the randomized training strategy was retained in the final model configuration.

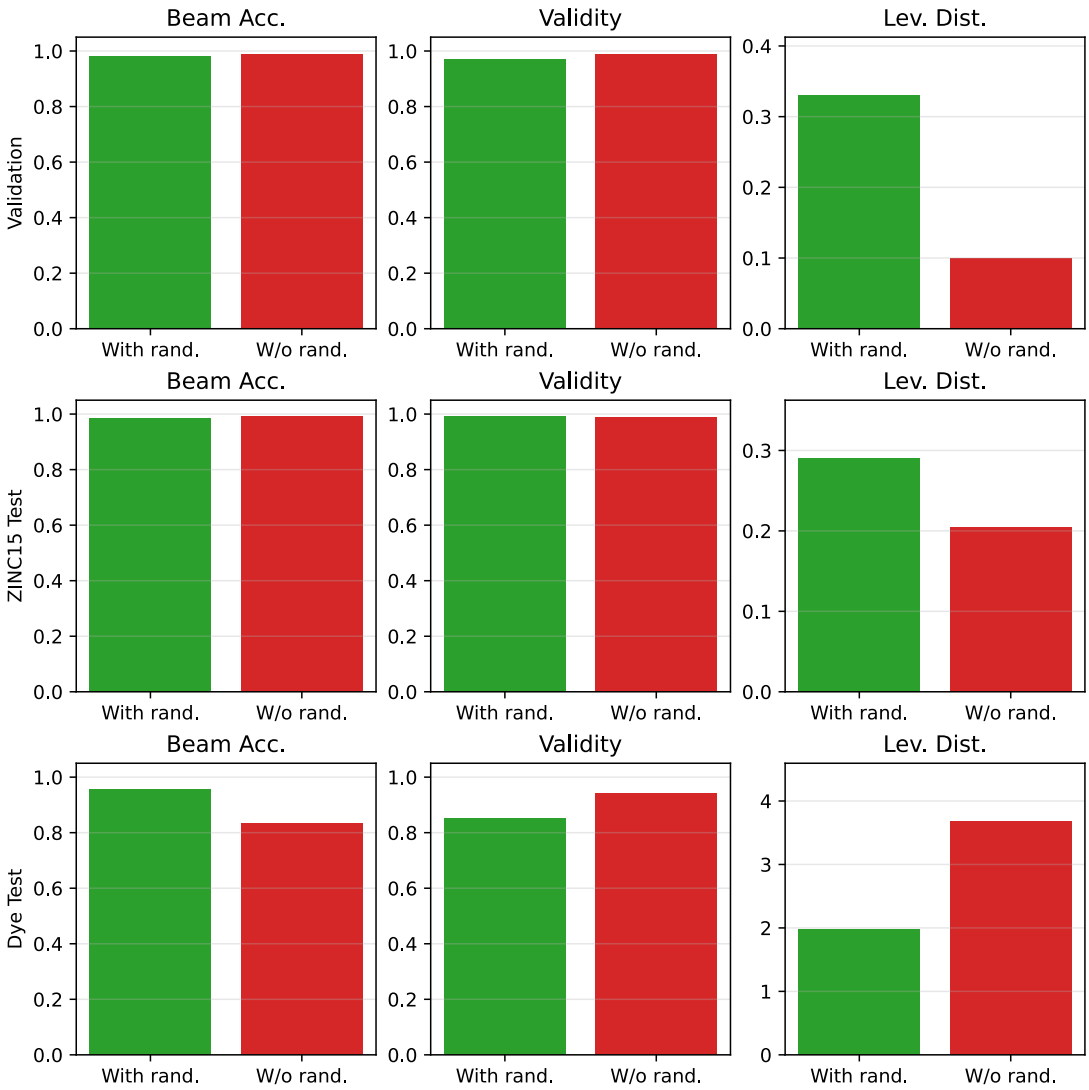


Figure S3. Ablation analysis of SMILES randomization and input regularization strategies.
Performance comparison of the proposed LTVAE model with and without randomized SMILES augmentation, token corruption, and label smoothing across three evaluation settings: (a) validation set, (b) ZINC15-like test set, and **(c)** dye SMILES test set. Each panel reports beam-search reconstruction accuracy, chemical validity ratio, and average Levenstein distance. While models trained without randomization achieve slightly higher reconstruction performance on in-distribution data, the inclusion of randomized SMILES and token-level corruption improves generalization to the structurally distinct dye dataset, yielding higher validity and reduced structural deviation. These results indicate that encoder-side SMILES randomization and decoder-side input corruption enhance robustness to out-of-distribution molecular structures.

REFERENCES

[1] Yoshikai, Y.; Mizuno, T.; Nemoto, S.; Kusuhara, H. A novel molecule generative model of VAE combined with Transformer. arXiv preprint arXiv:2402.11950 2024.

[2] Bhadwal, A. S.; Kumari, M.; Choudhary, S. TVAE: Transformer Based Variational Autoencoder for the Design of Drug Molecules. Proceedings of the International Conference on Computer, Electronics, Electrical, Engineering & their Applications (IC2E3). 2025.

[3] Gómez-Bombarelli, R.; Wei, J. N.; Duvenaud, D.; Hernández-Lobato, J. M.; Sánchez-Lengeling, B.; Sheberla, D.; Aguilera-Iparraguirre, J.; Hirzel, T. D.; Adams, R. P.; Aspuru-Guzik, A. Automatic chemical design using a data-driven continuous representation of molecules. ACS central science 2018, 4, 268–276.